

12.1.1 Phone Number Database Management

ALGORITHM

1. START
2. READ an integer n representing the total number of operations to perform.
3. INITIALIZE an empty dictionary named Ph.
4. FOR i from 1 to n :
 - READ the operation command and parameters as a list cmd.
 - IF cmd[0] is "ADD":
 - Add or update the contact: Ph[cmd[1]] = cmd[2].
 - ELSE IF cmd[0] is "REMOVE":
 - Remove the contact if it exists using Ph.pop(cmd[1], None).
 - ELSE IF cmd[0] is "DISPLAY":
 - IF the dictionary Ph is empty:
 - PRINT "No contacts".
 - ELSE:
 - SORT the dictionary keys (names) alphabetically.
 - FOR each name in the sorted keys:
 - PRINT the contact in the format {name}: {phone_number}.
5. END

CODETANTRA

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12.1.1. Phone Number Database Management

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

Input Format:

- First line contains an integer n representing the number of operations.
- Each of the next n lines contains a single operation in one of the following formats:
 - "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
 - "REMOVE <name>" - name is a string.
 - "DISPLAY" - no additional parameters.

Output Format:

- For display operation, print all contacts in alphabetical order by name in the format:
<name>: <phone_number>
- If the database is empty during display operation, print: "No contacts".
- Do not print anything for add and remove operations.

Constraints:

- Name contains only alphabetic characters (no spaces). And it is case sensitive.

Sample Test Cases

phoneDat...

```
1 n = int(input())
2 contacts = {}
3 for _ in range(n):
4     cmd = input().split()
5     if cmd[0] == "ADD":
6         name = cmd[1]
7         phone = cmd[2]
8         contacts[name] = phone
9     elif cmd[0] == "REMOVE":
10        name = cmd[1]
11        contacts.pop(name, None) # remove if exists, ignore otherwise
12    elif cmd[0] == "DISPLAY":
13        if not contacts:
```

Average time0.027 s27.13 msMaximum time0.049 s49.00 ms

5 out of 5 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 119 msDebug

Expected output

Actual output

4

ADD Alice 9876543210

ADD Bob 8765432109

ADD Charlie 7777777777

DISPLAY

Alice: 9876543210

Terminal

Test cases

< Prev

Reset

Submit

Next >

